
AGAIN?: INSIGHTS ON QUALITY-INFORMED DATA REPETITION FOR LANGUAGE MODEL PRE-TRAINING

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ABSTRACT

As models continue to scale and our supply of unique data dwindles, it becomes increasingly important to consider how we can make the most of the data we do have. Data filtering methods tell us that not all tokens are created equal. Scaling laws allow us to reliably predict performance across scales. Considering the tradeoff between data repetition and downstream performance, it has been shown that repeating data can be beneficial, but with diminishing returns. However, there has been little work considering the intersection of repetition and filtering; can a filtered multiset outperform a strict subset?

In this work, we approach this question, using different notions of quality and model performance. We find that in certain cases, we can maximize performance while using as much as 5x less unique data. In other instances, however repetition can result in disastrous performance, demonstrating the need to continue investigating this question as data becomes scarce.

Code can be found at https://github.com/camilobrownpinilla/CF_DCLM

1 Introduction

Large Language Models (LLMs) have proven to be one of the most rapidly evolving technologies of recent years. As demand for better models has grown, predictable performance patterns have allowed us to confidently invest in scaling these models and the datasets they are trained on, delivering remarkable performance in return [Hoffmann et al., 2022, Kaplan et al., 2020]. However, scaling has its limits; Villalobos et al. [2024] predict we will run out of fresh data to train models on by the year 2028. This, combined with the observation that models are only as good as the data they are trained on [Rae et al., 2022, Longpre et al., 2023], lends itself towards a natural question: *given a fixed data budget, how do we select an optimal subset to maximize downstream performance?*

In general, answering this question is difficult and computationally demanding. Regardless, recent research exploring this question has provided fruitful results. Simple, heuristic based approaches like the language model based Fineweb-Edu filter have created state-of-the-art open source datasets, producing models that excel on knowledge and reasoning based tasks [Penedo et al., 2024]. More complex, model-based methods like CoLoR-Filter produce datasets that can theoretically optimize for *any* downstream task, in some cases matching performance with up to 25x less data [Brandfonbrener et al., 2024]. Clearly then filtering is worth the effort. While different approaches yield different results for different use-cases, all are united by the implicit establishment of some notion of data quality. This commonality invites a subtler question: *given some notion of quality, just how much better are good tokens than bad ones? Can an optimal multiset be better than an optimal subset?*

Initial explorations of related questions have found that repeating examples can indeed be useful, but with diminishing returns [Muennighoff et al., 2023]. However, these investigations have been agnostic to any notion of quality. In this paper, we investigate how repeating the *best* tokens according to some notion of quality affects downstream performance on a variety of tasks.

2 Background

2.1 Fineweb-Edu Filter

Fineweb-Edu is a subset of the popular Fineweb dataset, created by filtering Fineweb for data deemed to be the most educational [Penedo et al., 2024]. To construct the filter, the authors prompted a language model to rate a selection of examples on their educational content on a scale of 1 to 5. Then, a linear regression model was trained on top of these examples and used to score the entire dataset of 15 trillion tokens, reducing it to a dataset of the 1.3 trillion most educational examples.

In the original paper, the authors found Fineweb-Edu to outperform all other public datasets by a significant margin [Penedo et al., 2024]. In particular, models trained on Fineweb-Edu excel on reasoning-based tasks. Fineweb-Edu is currently among the best performing datasets in the DataComp-LM (DCLM) competition, a benchmark designed to measure the quality of language-model pre-training datasets [Li et al., 2024].

2.2 CoLoR-Filter

Introduced by Brandfonbrener et al. [2024], CoLoR-Filter is a model based filtering algorithm that frames data selection as a Bayesian optimization problem. Given a large pre-training dataset D_{train} , a small "good" dataset sampled from downstream task(s) of interest D_{down} , CoLoR-Filter finds the fixed-size training set $S \subset D_{\text{train}}$, $|S| = n$ that minimizes the loss of target model θ on the downstream tasks D_{down} was sampled from. This is equivalent to selecting S that maximizes the likelihood of the downstream data, formulated by

$$\min_{S \subset D_{\text{train}}} -\log P(D_{\text{down}}|S)$$

Applying Bayes' rule and introducing a prior over θ , this is refined to

$$\min_{S \subset D_{\text{train}}} -\log \int_{\theta} P(S|\theta)P(\theta|D_{\text{down}}) + \log \int_{\theta} P(S|\theta)P(\theta)$$

Finally, after further conditioning and estimation, this is simplified to

$$\approx \min_{S \subset D_{\text{train}}} -\log(S|\theta_{\text{cond}}) + \log(S|\theta_{\text{prior}})$$

where θ_{prior} is an instance of θ trained on a subset $D_{\text{prior}} \subset D_{\text{train}}$ and θ_{cond} is trained on D_{prior} and D_{down} . In practice, θ_{cond} is initialized from θ_{prior} and finetuned on D_{down} . For more details on the derivation, we defer to the original paper [Brandfonbrener et al., 2024].

In short, CoLoR-Filter calculates a score for a datapoint based on the difference in loss between two auxiliary models, θ_{prior} and θ_{cond} . The selection algorithm proceeds by choosing the top n scoring points to form S . Crucially, CoLoR-Filter defines θ_{cond} , θ_{prior} to be the same model as θ . The utility of this method comes from scaling the size of θ while keeping θ_{cond} , θ_{prior} small, allowing for customizable, relatively cheap data selection.

2.3 DataComp-LM (DCLM)

LLMs are expected to perform well on a wide variety of diverse tasks when in deployment. To have any sense of the pace of progress of AI or the performance of a model, comprehensive evaluations are needed. Even still, there is little to no control when comparing across models. Models are often compared with different architectures, different hyperparameters, and different training regimens, making truly fair comparison nearly impossible.

To address this issue, Li et al. [2024] propose to instead compare datasets, creating a highly controlled competition environment where all other factors are held constant. Since the performance of models is highly dependent on the datasets they are trained on [Kaplan et al., 2020, Hoffmann et al., 2022], comparing the performance of datasets in place of the performance of models allows for a much more principled and robust measure of progress.

For our experiments, DCLM provides an ideal evaluation environment, allowing us to consider how data repetition informs model performance in a controlled and principled manner.

3 Proposed Approach: Two Notions, Two Testbeds

3.1 CoLoR Repetitions

We begin by investigating the effect of repeating the best data according to our first notion of quality: CoLoR scores, using DCLM as our evaluation environment.

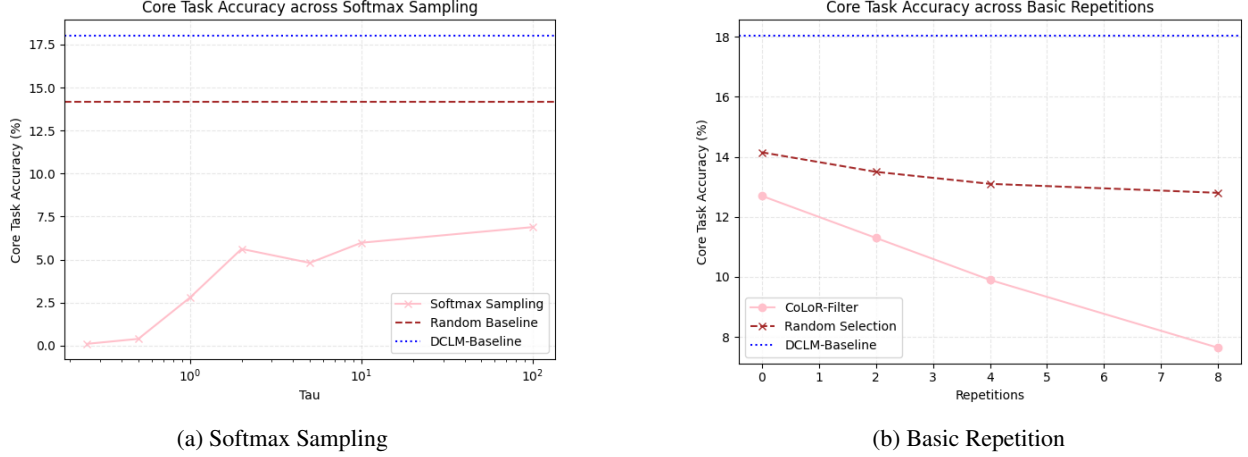


Figure 1: CoLoR-Filter selection schemes evaluated in DCLM.

Setup. We test how repeating different size subsets of the best data according to CoLoR scores impacts downstream performance, using DCLM as our evaluation environment. DCLM fixes the training recipe, allowing us to understand the performance of our datasets in isolation. There are various different tracks within DCLM, corresponding to different choices of model size. In our experiments, we train a 411M model on 8B tokens. DCLM requires that participants select from DCLM-Pool, their standardized corpus of 300T tokens.

Training CoLoR-Filter. We initialize $\theta_{\text{prior}}, \theta_{\text{cond}}$ using the OLMo codebase [Groeneveld et al., 2024]. We train θ_{prior} on DCLM-RefinedWeb, a lightly filtered subset of DCLM-Pool [Li et al., 2024]. The choice of dataset to finetune θ_{cond} on is crucial, but we currently do not have principled guidelines on how to select it. As a sane choice, we choose Core-Train-Tasks, a combination of all of the training sets of the 22 "core" evaluation tasks described by the DCLM authors.

Selecting data. We employ two different selection strategies, varying the amount of repetition within our 8B token budget for both methods.

Basic Repetition. Our first strategy is basic repetition. We select the top 8B tokens, select the top 4B and repeat twice, select the top 2B and repeat four times, and select the top 1B and repeat eight times.

Probabilistic Repetition. Our next strategy samples tokens for repetition according to their weight. Given scores $X = \{x_i\}^n$, we construct a probability distribution over scores with 'temperature' parameter τ via

$$P(x_i) = \frac{e^{x_i/\tau}}{\sum_{x_j \in X} e^{x_j/\tau}}$$

with tokens becoming equally likely as $\tau \rightarrow \infty$ and the highest scoring token becoming guaranteed as $\tau \rightarrow 0$. We sweep τ over $\{0.25, 0.5, 1, 2, 5, 10, 100\}$.

Baseline. As a baseline, we randomly sample 8B tokens, randomly sample 4B and repeat twice, and randomly sample 1B and repeat eight times. Additionally, we compare against the performance of DCLM-Baseline, a high-performing, baseline dataset released by the DCLM authors.

Evaluating DCLM. In addition to fixing the training recipe, DCLM fixes evaluation as well. Models are evaluated across a diverse set of 52 tasks, comprised of 22 "core" tasks and 30 "extended" tasks. Being that our CoLoR selections targets core tasks performance, we select aggregate performance on core tasks as our metric of interest.

3.2 Fineweb Repetitions

Our next set of experiments tackles the same question, but using the Fineweb-Edu filter as our notion of quality and OLMo as our testbed.

Setup. We test how different repetition schemes of the top tokens according to the Fineweb-Edu filter affects downstream performance. For our target model, we train 150M OLMo models, targeting the eight evaluation tasks in the original OLMo paper as our evaluation scheme [Groeneveld et al., 2024]. We select the Fineweb-Edu filter for study due to the

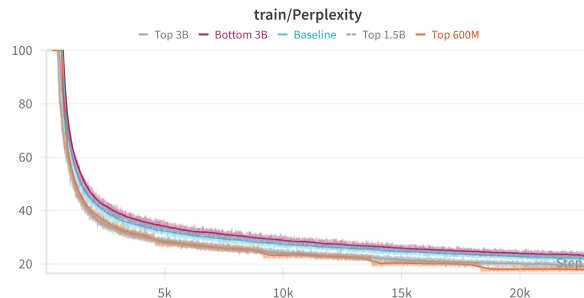


Figure 3: Train set perplexity for different Fineweb-Edu repetition schemes. Repeating the top 600M tokens 5 times achieves the highest performance.

very strong performance of Fineweb-Edu on DCLM, indicating the filter used to construct it may provide strong signal as to the quality of tokens. We train OLMo on 3B tokens in line with chinchilla optimality [Hoffmann et al., 2022].

Selecting data. We follow a simple repetition scheme, select top 3B tokens, the top 1.5B tokens and repeat twice, and the top 600M tokens and repeat six times. As a baseline, we compare the performance of 3B randomly sampled tokens. To better understand the global implications of token quality, we additionally consider the performance of the worst 3B tokens according to Fineweb-Edu.

Evaluating. To evaluate OLMo, we consider test accuracy on Hellaswag[Zellers et al., 2019], Winogrande [Sakaguchi et al., 2019], SciQ [Welbl et al., 2017], PiQA [Bisk et al., 2019], Openbook QA [Mihaylov et al., 2018], BoolQ [Clark et al., 2019], and Arc Easy and Arc challenge [Clark et al., 2018]. Additionally, we consider the perplexity [Jelinek et al., 1977] on the training set. While traditionally perplexity on a held-out test set is considered, we found that training perplexity produced a more meaningful signal, likely due to contamination in the test set.

3.3 A Note on Computational Cost

We recognize the need to consider a broad set of quality notions and repetition schemes. However, the cost of pre-training LLMs makes such a wide sweep prohibitively expensive.

4 Results

We first evaluate repeating the best selected CoLoR tokens in the DCLM environment (Figure 1). We observe an overall negative trend, with more repetition resulting in lower performance. Additionally, even training on the best 8B unique tokens results in performance that lags significantly behind DCLM-Baseline, and even behind random selection. Investigating further, we plot the distribution of CoLoR scores. We find that the mass of the distribution is overwhelmingly concentrated on negative scores (Figure 2)

Next, we consider repetition using Fineweb-Edu scores and OLMo as our evaluation environment. The Fineweb-Edu scoring model is a lightweight classifier. Additionally, we train OLMo models that are much smaller than DCLM models (150M vs. 411M), allowing for much cheaper and faster analysis. Plotting the perplexity on the training set, we observe the best performance with the most repetition (Figure 3). We then record the final accuracy on all downstream tasks (Table 1). We find no consistent winner, but see that the most repetition results in the highest accuracy for 3/8 tasks, and the second most repetition records the best accuracy for 4/8 other tasks. Averaging across across all tasks, we find that repeating the top 600M tokens five times results in the highest average accuracy of 51%, beating the baseline and bottom tokens by 0.86% and 3.51%, respectively.

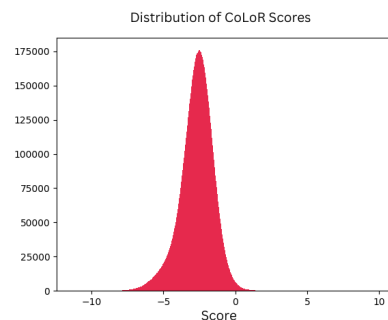


Figure 2: Distribution of CoLoR scores. Higher scores indicate better quality.

Table 1: Downstream Test Accuracy. **Best** in bold.

	Hellaswag	Winogrande	SciQ	PIQA	OpenBookQA	BoolQ	Arc Easy	Arc Challenge
Bottom 3B	38.3	51.2	71.5	63.1	30.8	51.7	48.5	24.8
Baseline	39.7	50.6	75.5	63.3	33.6	58.4	54.1	25.9
Top 3B	39.3	51.2	79.5	62.4	31.8	55.7	58.1	27.2
Top 1.5B	38.3	52.4	79.5	62.5	34.0	54.6	57.7	26.5
Top 600M	39.5	49.7	78.5	61.1	33.4	59.6	58.4	27.8

5 Discussion and Conclusion

Testing two different notions of quality with two different evaluation schemes, we find contradictory results. When using the scores produced by the Fineweb-Edu filter, we find that repeating the top 20% of tokens within the chinchilla compute budget results in the best performance overall, both when considering perplexity and downstream performance as metrics. In particular, we find that models trained on this data excel at tasks that are typically associated with reasoning and knowledge, such as BoolQ, Arc Easy and Arc Challenge. The good performance on this class of tasks in particular is likely due to the assignment of higher scores to more "educational" tokens.

When testing the effect of repeating CoLoR-Filtered data on DCLM, we observe contradictory results. We find negative trends in performance with increased repetition both when performing basic repetition and probabilistic sampling. Indeed, even when only training on the best unique tokens, performance significantly lags even behind random sampling. When constructing CoLoR-Filter, the choice of D_{down} on which to train θ_{cond} is crucial. However, it is generally difficult to understand which choice of D_{down} is best. Typically, choosing the training sets of the benchmarks you wish to optimize is a safe bet, but our results demonstrate that this is not the case. We note that part of this may be due to scale, as it is likely very difficult for a model of size 150M to fit to 22 different domains. However, the main benefit of CoLoR-Filter is the ability to use relatively small models to train models that can be orders of magnitude larger.

Overall, we have shown that repeating a small set of the best data several times can result in performance that beats even training on the best set of unique data— but that this result is highly dependent on the selected notion of data quality. These results open the door for further investigation of this problem across model scales, potentially allowing us to further maximize model performance even as data becomes scarce.

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